Given a permuation of the nodes $\boldsymbol{\sigma} = (\sigma_1, \ldots, \sigma_d)$ and some score vector $\boldsymbol{s} = (s_1, \ldots, s_d)$ for the nodes, we define the likelihood of the permutation under this score as

$$\log P_{PL}(\boldsymbol{\sigma}; \boldsymbol{s}) = \log P_{PL}\left(\sigma_1, \ldots, \sigma_d; \boldsymbol{s}\right) = \sum_{j=1}^{d} \left[\log s(\sigma_j) - \log \left(\sum_{k=j}^{d} s(\sigma_k) \right) \right] \tag{1}$$

Given a dataset $\mathcal{D} = \left\{ \left(X_1^{(i)}, \ldots, X_d^{(i)} \right) \right\}_{i=1}^{N}$ and a flow model f_θ, which takes a row of the data and a permutation as input and outputs the log-likelihood of observing that row under this permuation, we define the likelihood of $\mathcal{D}$ as follows:

$$\begin{aligned}
\log P(\mathcal{D}; \theta; \boldsymbol{s}) &= \sum_{i=1}^{N} \log P(X_1^{(i)}, \ldots, X_d^{(i)}; \theta; \boldsymbol{s}) && (2)\\
&= \sum_{i=1}^{N} \log \mathbb{E}_{\boldsymbol{\sigma} \sim P_{PL}(\cdot; \boldsymbol{s})} \left[P(X_1^{(i)}, \ldots, X_d^{(i)}; \theta; \boldsymbol{\sigma}) \right] && (3)\\
&= \sum_{i=1}^{N} \log \mathbb{E}_{\boldsymbol{\sigma} \sim P_{PL}(\cdot; \boldsymbol{s})} \left[\exp \left\{ f_\theta \left(X_1^{(i)}, \ldots, X_d^{(i)}; \boldsymbol{\sigma} \right) \right\} \right] && (4)
\end{aligned}$$

Now, let's derive the maximum-likelihood estimation of parameters $\{\theta, \boldsymbol{s}\}$. We run a gradient-based optimization to find the best parameters:

$$\begin{aligned}
\nabla_\theta \log P(\mathcal{D}; \theta; \boldsymbol{s}) &= \sum_{i=1}^{N} \nabla_\theta \log P(X_1^{(i)}, \ldots, X_d^{(i)}; \theta; \boldsymbol{s}) && (5)\\
&= \sum_{i=1}^{N} \nabla_\theta \log \mathbb{E}_{\boldsymbol{\sigma} \sim P_{PL}(\cdot; \boldsymbol{s})} \left[P(X_1^{(i)}, \ldots, X_d^{(i)}; \theta; \boldsymbol{\sigma}) \right] && (6)\\
&= \sum_{i=1}^{N} \frac{\mathbb{E}_{\boldsymbol{\sigma} \sim P_{PL}(\cdot; \boldsymbol{s})} \left[\nabla_\theta \exp \left\{ f_\theta \left(X_1^{(i)}, \ldots, X_d^{(i)}; \boldsymbol{\sigma} \right) \right\} \right]}{\mathbb{E}_{\boldsymbol{\sigma} \sim P_{PL}(\cdot; \boldsymbol{s})} \left[\exp \left\{ f_\theta \left(X_1^{(i)}, \ldots, X_d^{(i)}; \boldsymbol{\sigma} \right) \right\} \right]} && (7)\\
&= \sum_{i=1}^{N} \frac{\mathbb{E}_{\boldsymbol{\sigma} \sim P_{PL}(\cdot; \boldsymbol{s})} \left[\nabla_\theta f_\theta \left(X_1^{(i)}, \ldots, X_d^{(i)}; \boldsymbol{\sigma} \right) \cdot \exp \left\{ f_\theta \left(X_1^{(i)}, \ldots, X_d^{(i)}; \boldsymbol{\sigma} \right) \right\} \right]}{\mathbb{E}_{\boldsymbol{\sigma} \sim P_{PL}(\cdot; \boldsymbol{s})} \left[\exp \left\{ f_\theta \left(X_1^{(i)}, \ldots, X_d^{(i)}; \boldsymbol{\sigma} \right) \right\} \right]} && (8)
\end{aligned}$$

$$\begin{aligned}
\nabla_{\boldsymbol{s}} \log P(\mathcal{D}; \theta; \boldsymbol{s}) &= \sum_{i=1}^{N} \nabla_{\boldsymbol{s}} \log P(X_1^{(i)}, \ldots, X_d^{(i)}; \theta; \boldsymbol{s}) && (9)\\
&= \sum_{i=1}^{N} \nabla_{\boldsymbol{s}} \log \mathbb{E}_{\boldsymbol{\sigma} \sim P_{PL}(\cdot; \boldsymbol{s})} \left[\exp \left\{ f_\theta \left(X_1^{(i)}, \ldots, X_d^{(i)}; \boldsymbol{\sigma} \right) \right\} \right] && (10)\\
&= \sum_{i=1}^{N} \frac{\mathbb{E}_{\boldsymbol{\sigma} \sim P_{PL}(\cdot; \boldsymbol{s})} \left[\nabla_{\boldsymbol{s}} \log P_{PL}(\boldsymbol{\sigma}; \boldsymbol{s}) \cdot \exp \left\{ f_\theta \left(X_1^{(i)}, \ldots, X_d^{(i)}; \boldsymbol{\sigma} \right) \right\} \right]}{\mathbb{E}_{\boldsymbol{\sigma} \sim P_{PL}(\cdot; \boldsymbol{s})} \left[\exp \left\{ f_\theta \left(X_1^{(i)}, \ldots, X_d^{(i)}; \boldsymbol{\sigma} \right) \right\} \right]} && (11)
\end{aligned}$$

where,

$$\nabla_{\boldsymbol{s}} \mathbb{E}_{\boldsymbol{\sigma} \sim P_{PL}(\cdot;\boldsymbol{s})} \left[\exp\left\{ f_\theta \left(X_1^{(i)}, \ldots, X_d^{(i)}; \boldsymbol{\sigma} \right) \right\} \right] \tag{12}$$

$$= \nabla_{\boldsymbol{s}} \sum_{\boldsymbol{\sigma}} P_{PL}(\boldsymbol{\sigma}; \boldsymbol{s}) \cdot \left[\exp\left\{ f_\theta \left(X_1^{(i)}, \ldots, X_d^{(i)}; \boldsymbol{\sigma} \right) \right\} \right] \tag{13}$$

$$= \sum_{\boldsymbol{\sigma}} \nabla_{\boldsymbol{s}} P_{PL}(\boldsymbol{\sigma}; \boldsymbol{s}) \cdot \left[\exp\left\{ f_\theta \left(X_1^{(i)}, \ldots, X_d^{(i)}; \boldsymbol{\sigma} \right) \right\} \right] \tag{14}$$

$$= \sum_{\boldsymbol{\sigma}} P_{PL}(\boldsymbol{\sigma}; \boldsymbol{s}) \frac{\nabla_{\boldsymbol{s}} P_{PL}(\boldsymbol{\sigma}; \boldsymbol{s})}{P_{PL}(\boldsymbol{\sigma}; \boldsymbol{s})} \cdot \left[\exp\left\{ f_\theta \left(X_1^{(i)}, \ldots, X_d^{(i)}; \boldsymbol{\sigma} \right) \right\} \right] \tag{15}$$

$$= \sum_{\boldsymbol{\sigma}} P_{PL}(\boldsymbol{\sigma}; \boldsymbol{s}) \left(\nabla_{\boldsymbol{s}} \log P_{PL}(\boldsymbol{\sigma}; \boldsymbol{s}) \right) \cdot \left[\exp\left\{ f_\theta \left(X_1^{(i)}, \ldots, X_d^{(i)}; \boldsymbol{\sigma} \right) \right\} \right] \tag{16}$$

$$= \mathbb{E}_{\boldsymbol{\sigma} \sim P_{PL}(\cdot;\boldsymbol{s})} \left[\nabla_{\boldsymbol{s}} \log P_{PL}(\boldsymbol{\sigma}; \boldsymbol{s}) \cdot \exp\left\{ f_\theta \left(X_1^{(i)}, \ldots, X_d^{(i)}; \boldsymbol{\sigma} \right) \right\} \right] \tag{17}$$

Therefore, if we define our loss function as following:

$$\ell(\theta, \boldsymbol{s}) = \frac{1}{N} \sum_{i=1}^{N} \frac{\mathbb{E}_{\boldsymbol{\sigma} \sim P_{PL}(\cdot;\boldsymbol{s})} \left[\left(\log P_{PL}(\boldsymbol{\sigma}; \boldsymbol{s}) + f_\theta \left(X_1^{(i)}, \ldots, X_d^{(i)}; \boldsymbol{\sigma} \right) \right) \cdot sg \left(\exp\left\{ f_\theta \left(X_1^{(i)}, \ldots, X_d^{(i)}; \boldsymbol{\sigma} \right) \right\} \right) \right]}{\mathbb{E}_{\boldsymbol{\sigma} \sim P_{PL}(\cdot;\boldsymbol{s})} \left[sg \left(\exp\left\{ f_\theta \left(X_1^{(i)}, \ldots, X_d^{(i)}; \boldsymbol{\sigma} \right) \right\} \right) \right]} \tag{18}$$

where sg means stop-gradient.